

Sanjukta Das

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Work Permit - EAD Card

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<https://sanjukta-das.github.io/>

EDUCATION

M.S. in Computer Science (3.67/4)

Portland State University

 2018 – Present

Post-bacc in Computer Science (3.33/4)

Portland State University

 2017 – 2018

Computer Science (Transfer)

Portland Community College

 2016 – 2018

MBA in Finance

Regional College of Management, India

 2011 – 2013

WORK EXPERIENCE

IOTG Profit Optimization Intern

Intel Corporation, AZ

 June - December 2019

- Collaborate with multiple teams (OM Task Force, IOTG-Finance, SMG) for data collection, analysis, modeling & visualization.
- Built real-time dashboard for tracking year-round IOTG Operating Margin.
- Built interactive team website with one stop project management repository using SharePoint, HTML, CSS and JavaScript.
- Built executive view dashboard with SKU level margin tool for SMG team to set margin goals and allow comparable margin visibility.

HR Executive–US Payroll

HSBC HDPI, India

 2014 - 2015

BusinessDevelopmentOfficer–ModernTrade

Parle Agro, India

 2013

SKILLS

Technical : Linux, C, C++, Python , Flask, AWS, Google Cloud Platform, Docker, GIT, Bitbucket, Java, JUnit Testing, MongoDB, SQL, Google Web Toolkit, JSP, Servlet, REST API, Android, HTML, CSS, Javascript, SharePoint, Power BI, C#, .NET

Business : Ratio Analysis, Cash Flow Management, Security Analysis, Capital Budgeting, Trading & P/L, Balance Sheet Reports.

COURSE WORK

Technical : Full-Stack Web Development, Internet and cloud Systems, Cloud & Cluster database, Java Programming, Data Structure, Operating Systems, Linux, Software Engineering, Computational Photography, Algorithm, Database Management System, etc.

Business : Quantitative Techniques, Financial Services, Security Analysis, Capital Budgeting, Financial Planning and Wealth Management, etc.

ACADEMIC PROJECTS

Instapix (Flask, SQL, gcloud storage, Cloud Vision & Translate API): Designed and developed a Google cloud based application to allow users to follow other users, share/comment/like posts/photos, and view timeline etc.

Bubble Tea (Flask, SQL, GCP): Designed and developed a Google cloud based application to allow users to add different Bubble Tea store details (name, location, hours, contact, drinks, etc.) and give reviews.

Web Crawler (Python, BS4): traverse the world wide web to create search engine index with given seed links.

Operating Systems XV6(C, UNIX): Implemented **System calls:** a new system calls for date and time. **Process:** system calls to support and obtain process ownership, set and extract UID, GID, PID, track amount of time a process uses CPU. **Shell:** create new user command('ps' and 'ctrl+p' command), extract process detailed information. Scheduling: implemented MLFQ scheduler. **File system:** chmod, chown, chgrp system calls to set and modify file permissions that persist across xv6 sessions. Modified 'exec' and 'ls' command to enforce and display new file permissions respectively.

Buddy System (C): module that imitate malloc and free to manage a block of memory(e.g. heap)

Twitter (Java): An application which emulates the Twitter. Given a set of tweets as input, it extracts information such as date and time, user, tags, keywords, etc. and categorizes the tweets based on it.

Chat Application (Java): A Text-based messaging application using Java swing for designing GUI, HTML for aligned display of messages, Java socket for client-server communication, MongoDB to persist users information.

Phone Bill (Java, GWT): web application that maintains the phone call record details. Features implemented are: display a README manual describing how the program works, adding a new call record, display details of all calls or calls recorded within a certain period of time.

Computational Photography Projects (C++): Image processing **Toonify** - gives a cartoon like effect to images. Emulated cel-shading effects by applying median filter, canny edge detection, morphological operation, edge filtering and bilateral filter. **Mosaic** - gives a tiling effect, implemented by splitting the image to tiles, then averaging the color values and replacing each tile with a matching image. **Circle Packing** - gives the effect of cluster of circles by storing all the pixel values to fill the the circles and compare the (x, y) position within the image bounds to prevent overlapping of circles.

Hangman (Android): Developed a word guessing game given a limited number of wrong guesses. It selects and sets a random word from a text file and asks the user to guess the letters. Displays letters or next hanging picture on every correct or wrong guess respectively. The user loses once the hangman picture is complete.

Data Structure & Algorithms (Java and C++): Implemented **Multiply linked list:** with three separate link chains to keep the data entries unique and sorted by three different fields. **AVL Tree:** set container like in STL using AVL tree that supports union, intersection, cross product, and power set operations. **Sorting Algorithms:** analyzed various sorting algorithms using random data set. **Graph Algorithm:** Kruskal's and Prim's method of finding the Minimum Spanning Tree on random graphs, using union find data structure. **Shaunting Yard:** Dijkstra's shaunting yard algorithm using stack to evaluate logical expressions with AND, OR, NOT, etc. based on their order of precedence.

REFERENCES

Available on request.